

CHEMICAL CHANGES – OXIDATION & REDUCTION

KEY VOCABULARY

Oxidation: loss of electrons (or gain of oxygen).

Reduction: gain of electrons (or loss of oxygen).

Redox: a reaction where oxidation and reduction both happen.

OIL RIG: Oxidation Is Loss, Reduction Is Gain (of electrons).

COMMON MISCONCEPTIONS

~~—Oxidation always involves oxygen.~~

✓ Oxidation is loss of electrons — oxygen need not be present.

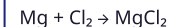
~~—Reduction means getting smaller.~~

✓ Reduction is gain of electrons / loss of oxygen.

~~—Only one species changes in redox.~~

✓ One is oxidised AND another is reduced.

REDOX EXAMPLE



Mg → **Mg²⁺ + 2e⁻** (oxidation — loses e⁻)

Cl₂ + 2e⁻ → 2Cl⁻ (reduction — gains e⁻)

Electrons lost = electrons gained.

Use OIL RIG to decide which is which.

1 DEFINE (ELECTRONS)

Define **oxidation** and **reduction** in terms of electrons. [2 marks]

2 IDENTIFY

In $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$, state whether zinc is oxidised or reduced. [1 mark]

3 HALF EQUATION

Write the **half equation** for chlorine gaining electrons. [2 marks]

4 OXYGEN VIEW

Copper oxide reacts with hydrogen. **Explain** which substance is reduced. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Oxidation is the ____ of electrons; ____ is the gain of electrons.

loss

gain

oxidation

reduction

6 REDOX OR NOT

Explain why a displacement reaction is a redox reaction. [2 marks]
